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PRN: B25ME1050 Div: SY3 Class: C3

INPUT:

```
#include <iostream>
using namespace std;

class Complex
{
public:

   int real, img;

   void accept(int r, int i)
   {
      real = r;
      img = i;
   }

   void display()
   {
      if (img >= 0)
         cout << real << "+i" << img;
      else
         cout << real << "-i" << -img;
   }

   Complex operator+(Complex c)
   {
      Complex x;
      x.real = real + c.real;
      x.img = img + c.img;
      return x;
   }

   Complex operator-(Complex c)
   {
```

```
    Complex x;
    x.real = real - c.real;
    x.img = img - c.img;
    return x;
  }

  Complex operator*(Complex c)
  {
    Complex x;
    x.real = real * c.real - img * c.img;
    x.img = real * c.img + img * c.real;
    return x;
  }

  Complex operator/(Complex c)
  {
    Complex x;
    double denom = c.real * c.real + c.img * c.img;

    x.real = (real * c.real + img * c.img) / denom;
    x.img = (img * c.real - real * c.img) / denom;

    return x;
  }
};

int main()
{
  Complex c1, c2, sum, diff, product, quotient;

  c1.accept(7, 6);
  c2.accept(5, 8);

  cout << "\nFirst Complex Number: ";
  c1.display();

  cout << "\nSecond Complex Number: ";
  c2.display();

  sum = c1 + c2;
```

```
    cout << "\nAddition of the complex numbers is: ";
    sum.display();

    diff = c1 - c2;
    cout << "\nSubtraction of these complex numbers is: ";
    diff.display();
    product = c1 * c2;
    cout << "\nMultiplication: ";
    product.display();


    quotient = c1 / c2;
    cout << "\nDivision: ";
    quotient.display();

    cout << endl;

    return 0;
}
```

OUTPUT:

Output

```
First Complex Number: 7+i6
Second Complex Number: 5+i8
Addition of the complex numbers is: 12+i14
Subtraction of these complex numbers is: 2-i2
Multiplication: -13+i86
Division: 0+i0

=== Code Execution Successful ===
```